

Glossary of Terms – Emerging Technologies

anything or everything as a service (XaaS)

– Array of cloud-based services or applications that are accessed on demand or on a subscription basis, including Software as a Service (SaaS), Infrastructure as a Service (IaaS), Platform as a Service (Paas) and others.

artificial intelligence (AI) – Using computers to apply advanced analysis and logic-based techniques to interpret events, support and automate decisions, and take actions; includes chat bots, machine learning, and robotics process automation.

automated fraud detection – Leverages emerging technologies to identify and prevent fraudulent activities in real-time.

autonomous and semi-autonomous things

– Self-managing technologies that operate independently or with limited human oversight to perform activities within a complex system including drones, smart-machine technologies, mobile robots, and autonomous or semi-autonomous vehicles.

business process integration tools – Tools that make independently designed systems work well together.

cloud computing services and solutions – Scalable and elastic IT-enabled capabilities delivered as a service using internet technologies.

cloud native solutions – Solutions and applications developed to take advantage of cloud computing.

collaboration tools – Software that enables collaboration through a shared network or online solutions including workstream and content collaboration tools with channels, chat features, audio/video calls, and shared documents.

connectivity (broadband/wireless/satellite connectivity)

– Ability to interconnect platforms and systems including enterprise networking, broadband, 5G services, and enhanced internet services.

data analytics solutions – Technologies for using data to inform planning, provide business intelligence, and enhance decision-making including predictive, prescriptive, and operationalized analytics.

data management tools – Technologies for data management strategies that put organizations in control of their business data, including the practice of classifying, storing, and terminating data.

distributed ledger technologies (including blockchain)

– A digital system with infrastructure, protocols, and software implementations providing a secure, verifiable ledger of transactions that can be simultaneously accessed, validated, and updated across multiple databases; a blockchain protocol is a subset of distributed ledgers that uses a specific data structure. Blockchain are transactional records shared by participants in a network that are cryptographically signed with irrevocable exchanges of value including virtual currency, digital assets, and smart contracts.

edge computing – A computing environment in which information processing, content collection, and delivery are placed closer to the sources and repositories of this information.

enterprise resource planning tools – The ability to deliver an integrated suite of business applications. ERP tools share a common process and data model, covering broad and deep operational end-to-end processes.

generative AI – Expands the output of AI systems to include the creation of high-value artifacts such as video, narrative, software code, synthetic data, and designs and schematics.

hyperautomation – Combination of tools such as AI, machine learning, and robotics process automation that the public sector can use to support enterprise initiatives for seamless access to public services.

low code/no code application development – Development platform that provides a development environment used to create application software through a graphical user interface instead of traditional hand-coded computer programming.

next generation application development – The development of scalable, reliant applications that meet broad business objectives using tools and techniques like low-code development platforms, containerization, and microservices.

open source software – Software operating under a more open licensing arrangement and built on a source code that is accessible and modifiable to all users.

quantum computing – A type of nonclassical computing that uses quantum bits (qubits) instead of binary 0's and 1's. Qubits can be linked, and quantum algorithms manipulate those linked qubits to solve complex problems exponentially faster than classical algorithms.

robotic process automation – A productivity tool that allows a user to configure one or more scripts (which some vendors refer to as “bots”) to activate specific keystrokes in an automated fashion.

security enhancement tools – Automated technology installed on endpoints to prevent and detect malicious actions, including user or entity behavioral analytics, multifactor authentication, endpoint detection and response, and host intrusion prevention or protection systems.

unified communications integration – Products including equipment, software, and services that provide and combine multiple enterprise communications channels, such as voice, video, personal and team messaging, voicemail, and content sharing.

virtual reality/augmented reality – Virtual reality is a computer-generated simulation or recreation of a real-life environment or situation. Augmented reality is computer generated enhancements to existing reality with ability to interact with it.

web3 – A new stack of technologies for the development of decentralized web applications that enable users to control their own identity and data. These technologies include blockchain as a trust verification mechanism, privacy-preserving and interoperability protocols, decentralized infrastructure and application platforms, decentralized identity, and support for applications like decentralized finance.